Safe Motion Chart - Joint Ranges

Heavy metal 4-DOF arm + gripper | Arduino Uno + PCA9685

File: docs/safe_motion_chart.md

Safe Motion Chart · Joint Ranges for Stable Pick-and-Place

Arm type: Heavy metal 4-DOF + gripper (MG995/MG996R joints, SG90 gripper)
Controller: Arduino Uno + PCA9685 (I2C, 50 Hz)
Purpose: Keep the arm stable, avoid servo overload and mechanical collision during pick-and-place.
Status: Starting safe ranges · must be calibrated per your physical arm (see Calibration).

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1. Joint Range Table (starting safe ranges)

Pulse values are PCA9685 ticks at 50 Hz (1 tick · 4.88 µs). Lower/higher limits must be calibrated with `joint_calibration` before use.

Joint	Channel	Servo	Safe Min (tick)	Safe Max (tick)	Neutral (tick)	Notes
Base (waist)	0	MG995	200	500	375	Keep sweep · ±60° from center to avoid cable twist
Shoulder	1	MG995	230	470	300	Heavy arm: avoid full horizontal extension under load
Elbow	2	MG995	220	480	350	Avoid fully straight (over-center) positions
Wrist pitch	3	MG90S/SG90	180	520	300	Keep gripper level during transport
Gripper	4	SG90	220 (open)	420 (grip)	·	Add 10·20 tick margin; do not stall

> **Rule:** Never command a joint to its absolute mechanical limit during motion. Keep a 10·20 tick margin.

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2. Pick-and-Place Waypoint Chart

All values in ticks. Move joints **sequentially** (base · shoulder · elbow · wrist · gripper) with 20 ms between joints to reduce peak current and collision risk.

Step	Base	Shoulder	Elbow	Wrist	Gripper	Delay (ms)	Purpose
0. Home (standby)	375	300	350	300	220 (open)	2000	Safe idle pose, arm elevated
1. Approach pick	250	420	280	250	220 (open)	1000	Hover above object
2. Grasp	250	420	280	250	420 (grip)	800	Close gripper
3. Lift	250	320	350	300	420 (grip)	1000	Clear pick zone
4. Transport to drop	500	320	350	300	420 (grip)	1000	Rotate to bin
5. Lower to bin	500	420	280	250	420 (grip)	800	Descend
6. Release	500	420	280	250	220 (open)	800	Open gripper
7. Return home	375	300	350	300	220 (open)	3000	Reset

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3. Safety Rules (heavy metal arm)

1. **Power:** XL4016 set to 5.0·6.0 V, ·6 A peak, common ground, 2200 µF cap on PCA9685.
2. **Never** power servos from Arduino 5 V pin.
3. **Calibrate first:** run `joint_calibration` per joint; update limits in code.
4. **Sequential motion:** move one joint at a time with 20 ms gaps (already in `moveArmSafe`).
5. **Payload limit:** ·180 g at full extension; reduce if arm droops.
6. **Gripper margin:** keep 10·20 tick gap from stall; test on soft objects first.
7. **No full-range sweeps** on the assembled metal arm · use calibrated safe ranges only.
8. **Mechanical stops:** add hard stops if a joint can over-rotate into the frame.
9. **Base clamping:** clamp the base to the table to prevent tipping.
10. **Dry run first:** 10 cycles with no payload before loaded test.

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4. Calibration Procedure (per joint)

1. Detach linkages; run `joint_calibration/joint_calibration.ino`.

2. Command tick 375 (neutral); attach horn at 90°.
3. Increment by 10 ticks until mechanical limit or hum/buzz; record as MAX.
4. Decrement to minimum; record as MIN.
5. Update `*_MIN/*_MAX` in `pick_and_place.ino` and `integrated_sorting.ino`.
6. Document in `tests/servo_test_results.md`.

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5. How to Use This Chart During Presentation

- Show Serial Monitor output (115200 baud) · each step prints its name.
- Run `pick_and_place` with no payload first; verify each joint stays in safe range.
- If any joint buzzes or stalls, stop immediately and reduce that joint's range.
- Keep the arm clamped to the table; stand clear during motion.

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6. Note on the Uploaded Arm Photo

Image received: `WhatsApp Image 2026-10-22 at 6.11.43 PM.jpeg` (899×1599 px).

I cannot visually inspect images, so please confirm:

- Which joint binds or collides first?
- Does the arm droop under its own weight?
- Does the gripper stall or fail to grip?
- Are any servo horns loose or slipping?

Your answers will let me tighten the safe ranges and waypoints further.